

»» Chapter I ««

THE REAL FIELD

PREFATORY REMARKS

It is taken for granted that the user of this book has already studied elementary real analysis. Desiderata for the purposes of complex function theory are: the properties of the real number system, limit processes, sequences, continuity, the Riemann-Stieltjes integral, and the basic theorems of the differential calculus in one and several variables. However, there is considerable variation of opinion concerning what elementary real analysis consists of. Also, the actual amount of material treated in any given class is apt to be very variable even in the presence of a fairly standard syllabus. A period of adjustment between the reader and the book is probably inevitable. In this chapter we give a succinct summary of material that should by and large be well known to the student. The familiar parts should be read for review, attention being given to the exercises. It is my experience that the notions of *limit superior* and *limit inferior* are much less commonly known by students than instructors might think. Since these notions are of considerable use in complex function theory, they merit prompt attention.

For readers wishing supplementary material we suggest Ch. 1 of [133], Chs. 1 and 2 of [36], and Chs. 1, 2, and 3 of [107].

1. SETS

The usual material of elementary set theory is so current that we may certainly take it for granted. Thus, without explanation we use as known the

notations $a \in A$, $A \subset B$, $A \cup B$, $A \cap B$. We also use $\emptyset$ for the empty set and $A - B$ for the set of elements in A but not in B . For the set specified by a sentential function $P(x)$, the set of x such that $P(x)$ holds, we write $\{x: P(x)\}$ and often the simpler $\{P(x)\}$. The universe of x in question will be clear from context. We also use curly brackets to indicate singletons and finite sets whose elements are exhibited explicitly. Thus, $\{a\}$ denotes the set whose sole member is a and $\{a, b\}$ denotes the set whose members consist exactly of a and b .

We take for granted the notion of an ordered pair (a, b) [following N. Wiener and Sierpiński we may define (a, b) as $\{\{a\}, \{a, b\}\}$] and that it has the property that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.† A *relation* is a set of ordered pairs. Given sets A and B , by the *cartesian product* of A and B , denoted $A \times B$, is meant the set of ordered pairs (a, b) satisfying $a \in A$, $b \in B$.

A *function* is a relation satisfying the condition that distinct ordered pairs belonging to it have distinct first components. Given a function f and $(a, b) \in f$, we denote b by $f(a)$. The set of the first components of the ordered pairs of a function is called the *domain* of the function, the second components the *values*, the set of second components the *image*. A function f is said to be *constant* provided that $f(x) = f(y)$ for all x, y in the domain of f . When the context renders the matter clear, we shall denote a constant f by the same notation as that used for its unique value when $f \neq \emptyset$. Given a function f with domain A and image contained in B , we say that f *maps* A *into* B and denote this assertion by $f: A \rightarrow B$.‡ If the image is B itself, we say that f *maps* A *onto* B . Given a function f and C , a subset of the domain of f , by the *restriction* of f to C , denoted $f|C$, is meant the set of ordered pairs belonging to f whose first components belong to C . Clearly, $f|C$ is a function. By the *image* of C with respect to f , denoted $f(C)$, is meant the image of $f|C$. We shall sometimes denote $f(C)$ by $\{f(x): x \in C\}$. Given $f: A \rightarrow B$, $D \subset B$, by the *preimage* of D with respect to f , denoted $f^{-1}(D)$, is meant $\{x: f(x) \in D\}$. A note of caution should be made for the case of numerical-valued functions: it is necessary to distinguish preimage from image with respect to the reciprocal of the given function. Suitable care will be taken notationally.

Composition. Given $f: A \rightarrow B$ and $g: B \rightarrow C$. By the *composition* of g with f , denoted $g \circ f$, is meant the set of ordered pairs $(x, g[f(x)])$, $x \in A$. It is a map of A into C .

Inverse. Given that f is a function. The set g of ordered pairs (b, a) , $(a, b) \in f$, is not necessarily a function.¶ In fact, g is a function if and only if

† Similarly we take for granted the notions of ordered triples (a, b, c) and ordered quadruples (a, b, c, d) .

‡ We shall also allow " $f: A \rightarrow B$ " to read " f mapping A into B ."

¶ g is the *inverse* of the relation f .

f is univalent [or in other words $(1, 1)$], that is, if $f(x) = f(y)$ implies $x = y$. In this case g is termed the *function inverse to f* , or more simply the inverse of f . It will be denoted by f^{-1} . It is to be noted that in the case of a univalent map of a set A onto a set B , the preimage of $C (\subset B)$ is the image of C with respect to f^{-1} .

Union and intersection of families of sets. Given a set A , by $\mathfrak{P}(A)$ we denote the set whose elements are the subsets of A , the set of *parts* of A . Given $\varphi: \Lambda \rightarrow \mathfrak{P}(A)$, by $\bigcup_{\lambda \in \Lambda} \varphi(\lambda)$ is meant the set of x belonging to $\varphi(\lambda)$ for some $\lambda \in \Lambda$ and by $\bigcap_{\lambda \in \Lambda} \varphi(\lambda)$ is meant the set of x belonging to $\varphi(\lambda)$ for all $\lambda \in \Lambda$. Commonly $\varphi(\lambda)$ is designated indicially by a notation such as X_λ .

Unions of functions. It may occur that each $\varphi(\lambda)$ is a function, say f_λ . It makes sense set-theoretically to consider $\bigcup_{\lambda \in \Lambda} f_\lambda$. For the purposes of complex function theory the operation of taking unions of functions is a particularly convenient means of constructing functions from “pieces.” For this reason the answer to the following question is wanted. When is the union a function? We leave for the reader the treatment of this question in Ex. 1.1 below.

EXERCISES

1.1. In the situation just considered a necessary and sufficient condition that $\bigcup_{\lambda \in \Lambda} f_\lambda$ be a function is that whenever a belongs to the domains of f_λ and f_μ , then $f_\lambda(a) = f_\mu(a)$.

1.2. The map ι of A onto itself given by $\iota(x) = x$, $x \in A$, is termed the *identity map of A onto itself*. Show that if f is a univalent map of A onto B , then $f^{-1} \circ f = \iota$ and $f \circ f^{-1}$ is the identity map of B onto itself.

A matter of notation. It frequently occurs that we specify a function f by declaring its domain A and the element $f(x)$ that f assigns to $x \in A$. One speaks of the function f “assigning $f(x)$ to $x \in A$ ” and more briefly (and inexactly) of “the function $f(x)$ with domain A ,” “the function $f(x)$, $x \in A$,” or simply “the function $f(x)$,” such licenses being taken when the facts are clear from context.

2. GROUPS, RINGS, FIELDS

We consider a set $G (\neq \emptyset)$ and a map $\cdot$ of $G \times G$ into G and we denote the image of (a, b) with respect to $\cdot$ by ab or $a \cdot b$. Such a map is termed a *binary operation* on G . The composite $(G, \cdot)$ is termed a *group* provided that (1) the *associative law* $(ab)c = a(bc)$, $a, b, c \in G$, holds, and (2) the equations $ax = b$ and $ya = b$, $a, b \in G$, admit solutions in G . It belongs to the very first lines of

group theory to show that the solutions are unique and further that there exists a unique element e (*neutral element, identity*) such that $ae = ea = a$, $a \in G$. The equations $ax = e$ and $ya = e$ have a common solution. It is termed the *inverse of a* and is denoted a^{-1} . It is easily seen that given $a \in G$ the map taking x into ax is a univalent map of G onto itself. The same holds for the map taking x into xa . The term "group" will be applied to G (the underlying set) as well as to $(G, \cdot)$. The elements of G are consequently called elements of the group. The tenor of this remark remains in force for all structures (algebraic and topological) which we shall meet.

If, in addition, the group operation is *commutative*, that is, $ab = ba$, $a, b \in G$, then the group is termed *abelian* (after the Norwegian mathematician N. H. Abel). In the case of abelian groups the additive notation is frequently (but not invariably) used, the group operation being written " $+$ " rather than " $\cdot$," the neutral element being termed *zero* and denoted 0 , the solution of $a + x = b$ being denoted $b - a$, and the inverse of a being denoted $-a$.

Given a group $(G, \cdot)$, a nonempty subset H of G is termed a *subgroup* provided that $(H, \cdot | H \times H)$ is a group. This is the case if and only if $ab^{-1} \in H$ whenever $a, b \in H$. If $A \subset G$, the intersection of all subgroups of G which contain A is itself a subgroup of G . It is termed the *subgroup of G generated by A* .

Groups of various kinds will appear in our study of complex function theory. One situation of interest is the following. Let A be a set and let G denote the set of univalent maps of A onto itself (*automorphisms of A*). We define the operation to be composition. Then $(G, \circ)$ is a group, the *group of automorphisms of A* . In concrete situations we frequently encounter subgroups of a group of automorphisms which are of particular interest. On several occasions in this book we shall meet such subgroups.

Given $A (\neq \emptyset)$ and binary operations $+$ and $\cdot$ on A , we say that the composite $(A, +, \cdot)$ is a *ring* provided that $(A, +)$ is an abelian group and that, in addition, (1) $\cdot$ is associative, and (2) the distributive laws $a(b + c) = ab + ac$, $(b + c)a = ba + ca$, $a, b, c \in A$, hold. If, in addition, $\cdot$ is commutative, A is termed a *commutative ring*. It is an easy consequence of the distributive laws that $0a = a0 = 0$, $a \in A$. The definitions for *subring* and the *subring generated* by a given subset of a ring parallel those for groups and need not be stated explicitly. The same observation holds for the following paragraph.

A commutative ring A is termed a *field* provided that A contains more than one element and that the equation $ax = b$, $a (\neq 0)$, $b \in A$, admits a solution (necessarily unique) in A . The solution is denoted b/a . A field A contains a unique element $1 (\neq 0)$ satisfying $1a = a$, $a \in A$. It is called the *identity (element)* of A . If $a, b \in A$, then $ab = 0$ if and only if at least one of the elements a, b is 0 . If $a \in A$ and $a \neq 0$, then the unique solution of $ax = 1$ is not 0 . It is termed the *reciprocal of a* and is denoted by a^{-1} as well as by $1/a$.

We note that $A - \{0\}$ together with $\cdot$ restricted to the cartesian product of $A - \{0\}$ with itself is an abelian group.

3. ORDERED FIELDS

Given a field A , $P(\subset A)$ is said to be a *positive part* of A provided that (1) $P \times P$ is mapped into P by both $+$ and $\cdot$ and (2) the sets $\{0\}$, P , and $\{x: -x \in P\}$ are disjoint and have A as their union. By an *ordered field* is meant a composite $(A, +, \cdot, P)$ where $(A, +, \cdot)$ is a field and P is a positive part of A . We are concerned for the remainder of this section with an ordered field A . An element p is termed *positive* provided that $p \in P$; an element n is termed *negative* provided that $-n \in P$. With the aid of the distributive law it is seen that $-(ab) = a(-b)$ and $ab = (-a)(-b)$. It follows readily that if p is positive and n_1, n_2 are negative, then pn_1 is negative and n_1n_2 is positive. Noting that $1 = (1)(1) = (-1)(-1)$, we see that 1 is positive.

EXERCISE

3.1. Prove the following property of an ordered field: Given elements a and b , then $a^2 + b^2$ is positive or 0. It is 0 if and only if $a = b = 0$.

Inequality. Given $a, b \in A$, we say that a is *less than* b , denoted $a < b$, provided that $b - a \in P$. We say that a is *greater than* b , denoted $a > b$, provided that $b < a$.† It is elementary that the law of *trichotomy* holds: Given $a, b \in A$, then one and only one of the following is valid: $a = b$, $a < b$, $b < a$. We write " $a \leq b$ " to denote " $a < b$ or $a = b$," " $a \geq b$ " to denote " $a > b$ or $a = b$." The elementary rules of inequalities follow readily by use of the definitions just put down and the notion of a positive part. We give some of the principal rules in the exercises that follow. It is to be understood that $a, b, c, d \in A$, p is positive, n is negative.

EXERCISES

3.2. $a > 0$ if and only if a is positive. $a < 0$ if and only if a is negative.

3.3. If $a < b$, then $a + c < b + c$.

3.4. If $a < b$ and $c < d$, then $a + c < b + d$.

† It would be better to use terms like "precede" and "follow," but there is no serious need to break with convention.

- 3.5. What is the corresponding conclusion when $a \leq b$ and $c \leq d$?
- 3.6. If $a \geq 0$, $b \geq 0$, and $a + b = 0$, then $a = b = 0$.
- 3.7. If $a < b$ and $b < c$, then $a < c$. What is the corresponding conclusion when $a \leq b$ and $b \leq c$? Show that if $a \leq b$, $b \leq c$, $a = c$, then $a = b$.
- 3.8. If $a < b$, then $pa < pb$ and $na > nb$.
- 3.9. If $0 \leq a < b$ and $0 \leq c < d$, then $0 \leq ac < bd$. Consider on the basis of your experience what possibilities may occur in the real number system when the conditions $0 \leq a$ and $0 \leq c$ are dropped.
- 3.10. If $0 < a < b$, then $0 < b^{-1} < a^{-1}$.
- 3.11. Show that $a^2 + b^2 \geq 2ab$. When does the case of equality occur?

The use of inequalities is extremely frequent in analysis and it is well to secure a firm control early. The cautionary note of Ex. 3.9 is to be heeded. For an authoritative systematic account of inequalities we cite [50].

We continue our review of notions that arise in the study of an ordered field and consider $X \subset A$. An element $a \in A$ is termed an *upper bound* of X provided that $a \geq x$, $x \in X$; a is termed a *lower bound* of X provided that $a \leq x$, $x \in X$. The set X is said to be *bounded above* provided that there exists $a \in A$ which is an upper bound of X . A corresponding definition is made with "above" and "upper" replaced by "below" and "lower," respectively. We say that X is *bounded* provided that X is both bounded above and bounded below. There is at most one element in X which is an upper bound of X . When such an element exists, it is termed the *maximum* of X and is denoted $\max X$. A corresponding assertion holds with "lower bound" replacing "upper bound" and we arrive at the notion of the *minimum* of a set ($\min X$). Given X , we consider the set Y of upper bounds of X and the set Z of lower bounds of X . When Y has a minimum, we term $\min Y$ the *supremum* of X and denote it $\sup X$; when Z has a maximum, we term $\max Z$ the *infimum* of X and denote it $\inf X$. It is to be noted that the terms "least upper bound" (l.u.b.) and "greatest lower bound" (g.l.b.) are employed commonly in English-speaking countries for "supremum" and "infimum," respectively. If X possesses a maximum, then $\max X = \sup X$. Similarly, if X possesses a minimum, then $\min X = \inf X$. To say that $\sup X = +\infty$ signifies that X is not bounded above and to say that $\inf X = -\infty$ signifies that X is not bounded below. These formal notations will be seen to go into meaningful assertions when X is a subset of the real number system and one considers the extended real line (§6, Ch. III). It is to be emphasized that, given $a \in A$, $X \subset A$, when we write " $\sup X = a$," it is implicit that the above Y does have a minimum. Cognate remarks are to be in force when we deal with notions such as "limit" and "limit superior."

The set $\{a, b\}$ has a maximum and a minimum as follows at once from the alternatives: $a \leq b$, $a > b$. From the point of view of analysis it is convenient to introduce a^+ , the *positive part* of a , defined as $\max\{a, 0\}$; a^- , the *negative part* of a , defined as $\min\{a, 0\}$; $|a|$, the *absolute value* of a ,

defined as $\max\{a, -a\}$. It is also convenient to introduce $A^+ = \{x \geq 0\}$ and $A^- = \{x \leq 0\}$.

EXERCISES

3.12. Verify the following:

$$\begin{aligned} a &= a^+ + a^-, & |a| &= a^+ - a^-, & -|a| &\leq a^- \leq a \leq a^+ \leq |a|, \\ (a+b)^+ &\leq a^+ + b^+, & |ab| &= |a||b|, & |a+b| &\leq |a| + |b|. \end{aligned}$$

When is $|a+b| = |a| + |b|$?

3.13. Determine the maps $f: A \rightarrow A$ satisfying $|f(x) - f(y)| = |x - y|$, $x, y \in A$. Show that the set of such maps together with the operation of composition constitutes a group.

Intervals. Given $a < b$, we define the *interval* $]a, b[$ as $\{a < x < b\}$ and the *semi-intervals* $]a, b]$ and $[a, b[$ as $\{a < x \leq b\}$ and $\{a \leq x < b\}$, respectively. Given $a \leq b$, we define the *segment* $[a, b]$ as $\{a \leq x \leq b\}$. We are following here the notation employed by Dieudonné [36].

4. THE REAL NUMBER SYSTEM

The theater of action of elementary classical analysis is the real number system. Our object in this chapter is to recall in an axiomatic way the basic facts that we shall want concerning the real number system. It is not our intention to motivate the axioms used nor to describe the historical evolution of the real number system and its constructive treatment such as that presented by Landau [71]. It is taken for granted that the reader has prior experience with a treatment of the real number system. We remark that the axiomatic approach to the real number system made an early appearance in Carathéodory's treatise on functions of a real variable [23].

We continue with an ordered field A . A is said to be *complete* provided that every nonempty $X (\subset A)$ which is bounded above possesses a supremum (caution: not necessarily in X). When we speak of the *real number system*, we mean a complete ordered field, the definite article of tradition notwithstanding. We shall reserve $\mathbf{R}$ to denote the real number system and its underlying set and $\mathbf{P}$ to denote the set of positive members of $\mathbf{R}$.

It is easy to see that the completeness property implies that a nonempty subset of $\mathbf{R}$ which is bounded below possesses an infimum.

We take advantage of the structural richness of $\mathbf{R}$ to introduce various important substrata.

$\mathbf{N}$, the set of natural numbers. We consider subsets X of $\mathbf{R}$ satisfying: (1) $0 \in X$, (2) $s(X) \subset X$ where $s: \mathbf{R} \rightarrow \mathbf{R}$ is given by $s(x) = x + 1$. We define $\mathbf{N}$ to be the intersection of such X and term its members the *natural* (or *whole*) numbers. $\mathbf{N}$ is the “smallest” X satisfying (1) and (2). Clearly, $\mathbf{N} \subset \mathbf{R}^+$. A number of important properties of $\mathbf{N}$ are given in the exercises that follow.

EXERCISES

4.1. $\mathbf{N}$, 0, and s satisfy the Peano axioms; that is, (a) s is a univalent map of $\mathbf{N}$ into $\mathbf{N} - \{0\}$, (b) if $X \subset \mathbf{N}$ satisfies $0 \in X$ and $s(X) \subset X$, then $X = \mathbf{N}$. Here (b) is just the induction property.

4.2. $\{0\} \cup s(\mathbf{N}) = \mathbf{N}$.

4.3. Let $x, y \in \mathbf{N}$. Then $x + y, xy \in \mathbf{N}$. When $x > y$, then $x - y \in s(\mathbf{N})$.

4.4. If $x, y \in \mathbf{N}$ and $x < y \leq x + 1$, then $y = x + 1$.

4.5. If $X (\neq \emptyset) \subset \mathbf{N}$, then X possesses a minimum (well-ordering property of $\mathbf{N}$).

4.6. Given $a, b \in \mathbf{R}$, $a > 0$. Then there exists $n \in \mathbf{N}$ such that $na > b$ (archimedian-ordering property of $\mathbf{R}$). Hint: $\{na: n \in \mathbf{N}\}$ is not bounded above, otherwise it would have a supremum, say α . Complete the argument noting that $\alpha - a$ is not an upper bound of the set being considered.

When we say that $P(n)$ holds for (*sufficiently*) *large* n , $P(n)$ being a sentential function defined for $n \in \mathbf{N}$, we mean that the set of $n \in \mathbf{N}$ for which $P(n)$ does not hold is bounded. When we say that $P(n)$ holds for *arbitrarily large* n , we mean that the set of $n \in \mathbf{N}$ for which $P(n)$ holds is unbounded.

For “ $k \in \mathbf{N}$ ” we write on occasion “ $k = 0, 1, \dots$ ”

By the *initial section* $\langle 0, n \rangle$ of $\mathbf{N}$, $n \in \mathbf{N}$, is meant $\{x \in \mathbf{N}, x \leq n\}$. Given $m, n \in \mathbf{N}$, by the *section* $\langle m, m + n \rangle$ of $\mathbf{N}$ is meant

$$\{x \in \mathbf{N}, m \leq x \leq m + n\}.$$

A set X is said to be *finite* provided that either $X = \emptyset$ or there exists $n \in \mathbf{N}$ such that X admits a univalent map onto $\langle 0, n \rangle$. There is at most one such n . $\langle 0, -1 \rangle$ will denote $\{x \in \mathbf{N}, x \leq -1\} = \emptyset$.

By a *sequence* we understand primarily a function with domain $\mathbf{N}$ and we shall denote such a sequence by notation such as $(a_n)_{n \in \mathbf{N}}$ or simply (a_n) . The image of $m \in \mathbf{N}$ with respect to (a_n) is called the *mth term* (or term of index m) of (a_n) and is denoted a_m (indicial notation) and also $a(m)$. Corresponding understandings hold for sequences in general. We also apply the term *sequence* to functions having certain preferred domains such as an initial section of $\mathbf{N}$. When the domain is the initial section $\langle 0, n \rangle$, to denote the sequence we employ a notation such as $(a_k)_0^n$. Given a sequence f with domain $\mathbf{N}$, by a *subsequence* of f is meant a sequence of the form $f \circ v$, where $v: \mathbf{N} \rightarrow \mathbf{N}$ is increasing, that is, $v(m) < v(n)$, $m, n \in \mathbf{N}$, $m < n$.

Inductive definition of sums and products. Given $(a_k)_0^n, a_k \in G$, a group. There exists a unique $(p_k)_0^n$ such that $p_0 = a_0, p_{k+1} = p_k a_{k+1}, k \in \mathbb{N}, k \leq n-1$ (for which we shall write, as is customary, “ $k = 0, \dots, n-1$ ”). We denote p_n by $\prod_0^n a_k$ and term it the *product* associated with $(a_k)_0^n$. We suppose, of course, that the reader is familiar with the usual rules for products and powers. For the abelian case additively denoted “ $\sum$ ” (*sum*) replaces “ $\prod$.” We note the conventions: $\prod_0^{-1} a_k = e$ (neutral element) and the abelian version $\sum_0^{-1} a_k = 0$.

EXERCISE

4.7. Abel’s identity. Given $(a_k)_0^n$ and $(b_k)_0^n$ where the elements a_k and b_k are members of a ring A . Let $s_k = \sum_0^k b_j$. Then

$$\sum_0^n a_k b_k = \left[\sum_0^{n-1} (a_k - a_{k+1}) s_k \right] + a_n s_n. \quad (4.1)$$

Note that the commutativity of multiplication is not used. This simple identity has many applications.

$\mathbb{Z}$, the set of integers. We define $\mathbb{Z}$ as the subgroup of $(\mathbb{R}, +)$ generated by $\{1\}$. We term the members of $\mathbb{Z}$ *integers*. The properties of $\mathbb{Z}$ which we shall want are given in the following exercises.

EXERCISES

4.8. $\mathbb{Z} = \mathbb{N} \cup \{x: -x \in \mathbb{N}\}$.

4.9. If $x, y \in \mathbb{Z}$, then $xy \in \mathbb{Z}$. [$\mathbb{Z}$ is closed with respect to multiplication.]

Sequences will, more generally, be understood to be functions with domain $\mathbb{Z}$ or a subset of one of the following forms: $\{x \in \mathbb{Z}, x \geq n\}$, $\{x \in \mathbb{Z}, x \leq n\}$, $\{x \in \mathbb{Z}, m \leq x \leq n\}$, where $m, n \in \mathbb{Z}, m \leq n$. In the final case we speak of a *finite sequence*. We shall use in the respective cases notations such as $(a_k)_{k \in \mathbb{Z}}$ or simply (a_k) , $(a_k)_{k \geq n}$, $(a_k)_{k \leq n}$, $(a_k)_m^n$. In the last case we also use $(a_k)_{m \leq k \leq n}$. Given $(a_k)_m^n$ where the a_k are members of a group, we define $\prod_m^n a_k$ as $\prod_0^{n-m} a_{m+k}$. A corresponding definition holds for the additive situation. The convention $\prod_m^{m-1} a_k = e$ (neutral element) will prevail as will the corresponding $\sum_m^{m-1} a_k = 0$. Continuing with m and n as above we term $\{x \in \mathbb{Z}, m \leq x \leq n\}$ a *section* of $\mathbb{Z}$ and denote it by $\langle m, n \rangle$. Also for “ $k \in \langle m, n \rangle$ ” we write on occasion “ $k = m, \dots, n$,” and for “ $k \in \mathbb{Z}, k \geq n$,” we write “ $k = n, n+1, \dots$ ”

Q, the set of rationals. We define **Q** as the intersection of the subfields of **R**. It is simply the subfield of **R** contained in all subfields of **R**. The members of **Q** are termed *rational numbers*.

Of the exercises that follow the first relates **Z** and **Q**. The others treat properties of **R**.

EXERCISES

4.10. $\mathbf{Q} = \{m/n : m, n \in \mathbf{Z}, n \neq 0\}$.

4.11. Given $a, b \in \mathbf{R}$, $a < b$. There exists $q \in \mathbf{Q}$ such that $a < q < b$ (density of rationals). Hint: It suffices to consider $a \geq 0$. [Why?] Let n denote the least natural number satisfying $n(b - a) > 1$. Let m denote the least natural number such that $m(n^{-1}) > a$. Then $q = m/n$ serves.

4.12. If $a \in \mathbf{R}$, then

$$a = \sup\{q \in \mathbf{Q}, q \leq a\} = \sup\{q \in \mathbf{Q}, q < a\}.$$

4.13. Intermediate value property of continuous functions. Given $a, b (> a) \in \mathbf{R}$. Let f be a continuous map of $[a, b]$ into $\mathbf{R}$; that is, it is supposed that for each $x \in [a, b]$ and each positive η there exists δ positive such that $|f(y) - f(x)| < \eta$ when $|y - x| < \delta$. Suppose that $f(a) < c < f(b)$. Then there exists $d \in]a, b[$ such that $f(d) = c$. Prove this well-known theorem of elementary analysis in the following way. Introduce $\{f(x) \leq c\}$ and show that $d = \sup\{f(x) \leq c\}$ is actually $\max\{f(x) = c\}$.

4.14. Given $n \in \mathbf{N}$, and a_k , $k = 0, \dots, n$, satisfying $a_k \in \mathbf{R}^+$, $\sum_0^n a_k > 0$. Show with the aid of Ex. 4.13 that there is a unique positive real number x satisfying $x^{n+1} - \sum_0^n a_k x^k = 0$. Hint: Consider $1 - \sum_0^n a_k x^{k-(n+1)}$ for $x > 0$.

Apply this result to show that if $a \in \mathbf{R}^+$ and n is a positive natural number, then there exists a unique $x \in \mathbf{R}^+$ such that $x^n = a$. We use the notations $a^{1/n}$ and $\sqrt[n]{a}$ ($\sqrt{a}$ when $n = 2$) for the unique x in question.

4.15. Let $f: \mathbf{R} \rightarrow \mathbf{R}$, not constant, satisfy

$$f(x + y) = f(x) + f(y) \quad \text{and} \quad f(xy) = f(x)f(y), \quad x, y \in \mathbf{R}.$$

Show that f is the identity map of $\mathbf{R}$ onto itself. Hint: Show that $f(x) = x$ on $\mathbf{N}$, $\mathbf{Z}$, $\mathbf{Q}$ successively and exploit the fact that if $a > 0$, then $b^2 = a$ for some $b \in \mathbf{R}$ to conclude that if $x < y$, then $f(x) < f(y)$. [The result of the exercise says that the only isomorphic map of $\mathbf{R}$ into itself is the identity map.]

4.16. Determine the maps $f: \mathbf{R} \rightarrow \mathbf{R}$ satisfying: (1) $f(x + y) = f(x) + f(y)$, $x, y \in \mathbf{R}$, (2) f is continuous at some $a \in \mathbf{R}$; that is, for each positive η there exists δ positive such that $|f(x) - f(a)| < \eta$ when $|x - a| < \delta$.

4.17. Suppose that A is a field and that P_1 and P_2 are positive parts rendering A a complete ordered field. Show that $P_1 = P_2$.

5. REAL SEQUENCES, MONOTONE SEQUENCES

The sequences in this and the following two sections are assumed to have domain $\mathbf{N}$ and take values in $\mathbf{R}$. We suppose that the reader has encountered

the usual elementary material on real sequences, that is, sequences with image in $\mathbf{R}$. Thus, we take for granted familiarity with the definition of *limit of a sequence* which follows:

Given a real sequence (a_n) and $\alpha \in \mathbf{R}$, we say that (a_n) has *limit* α provided that for each positive η there exists $m \in \mathbf{N}$ such that $|\alpha - a_n| < \eta$ for $n \geq m$.

In addition, we assume familiarity with such matters as *uniqueness* of limit; that is, if (a_n) has limit β also, then $\alpha = \beta$; the usual notation $\lim_{n \rightarrow \infty} a_n$ (also $\lim a_n$); the definitions for *sums*, $(a_n) + (b_n) = (a_n + b_n)$, and *products* $(a_n)(b_n) = (a_n b_n)$ endowing the set of real sequences with the structure of a commutative ring; and the associated limit theorems for sums, differences, products, and quotients. Possibly less familiar but no less immediate is the result that if (a_n) has limit α , then every subsequence of (a_n) has limit α , which follows from the definition of limit and the fact that if $\nu: \mathbf{N} \rightarrow \mathbf{N}$ and is increasing, then $\nu(n) \geq n$, $n \in \mathbf{N}$.

The notion of a limit may be viewed in the following manner. Let $\mathfrak{L}$ denote the set of ordered pairs $((a_n)_{n \in \mathbf{N}}, \alpha)$ where (a_n) is a real sequence, $\alpha \in \mathbf{R}$, and (a_n) has limit α . Then $\mathfrak{L}$ is a *function* and $\lim a_n$ is just $\mathfrak{L}((a_n))$. It might be well to note that *implicit* in taking for granted that “ $\lim a_n$ ” is meaningful is the assumption that (a_n) belongs to the domain of $\mathfrak{L}$, that is, that for some $\alpha \in \mathbf{R}$ the sequence (a_n) has limit α . These remarks hold elsewhere in the study of limits. Corresponding observations hold for limit superior and limit inferior (§6, Ch. I).

In this section we consider the existence of limits of monotone sequences. A real sequence (a_n) is said to be *increasing* provided that $a_m < a_n$ whenever $m < n$. It is said to be *never decreasing* provided that $a_m \leq a_n$ whenever $m < n$. Corresponding definitions are made for *decreasing* ($a_m > a_n$ whenever $m < n$) and *never increasing* ($a_m \geq a_n$ whenever $m < n$). We write “ $(a_n)^\uparrow$ ” to denote “ (a_n) never decreasing” and “ $(a_n)^\downarrow$ ” to denote “ (a_n) never increasing.” A sequence is termed *monotone* provided that it is either never decreasing or never increasing.

We consider $(a_n)^\uparrow$. There are two possibilities to consider. One is that $\{a_n: n \in \mathbf{N}\}$ is not bounded above. Here for each $b \in \mathbf{R}$ there exists m such that $a_m > b$ and consequently $a_n > b$ for $n \geq m$. Hence $\lim_{n \rightarrow \infty} a_n = +\infty$.† The other possibility is that $\{a_n: n \in \mathbf{N}\}$ is bounded above. By the completeness property of $\mathbf{R}$, the set $\{a_n: n \in \mathbf{N}\}$ has a supremum α . Given positive η , $\alpha - \eta$ is not an upper bound of $\{a_n: n \in \mathbf{N}\}$. Hence $a_m > \alpha - \eta$ for some $m \in \mathbf{N}$ and consequently

$$\alpha - \eta < a_n \leq \alpha$$

for $n \geq m$. We conclude that $\alpha = \lim_{n \rightarrow \infty} a_n$.

† We recall that a real sequence (a_n) is said to have limit $+\infty$, denoted “ $\lim_{n \rightarrow \infty} a_n = +\infty$,” provided that for each real b there exists $m \in \mathbf{N}$ such that $a_n > b$, $n \geq m$.

Hence a monotone never decreasing real sequence either has limit $+\infty$ or else has limit α for some $\alpha \in \mathbf{R}$.† Obviously, a corresponding statement holds for $(a_n)_{\downarrow}$.

EXERCISES

5.1. Following A. Hurwitz [59; pp. 706–721] we may establish the existence of the square root of a positive real number on the basis of the theorem just treated concerning the limit of a monotone sequence and, in fact, conclude the existence of an increasing function L on $\mathbf{P}$ satisfying

$$L(xy) = L(x) + L(y), \quad x, y \in \mathbf{P}. \quad (5.1)$$

Given $a > 1$, we consider the sequence (a_n) satisfying

$$a_0 = a, \quad a_{n+1} = (a_n + a_n^{-1}a)/2, \quad n \in \mathbf{N}.$$

Show that (a_n) is decreasing and that $\alpha = \lim a_n > 1$ and $\alpha^2 = a$. Conclude the existence of a positive square root for positive a .

Continuing with $L_n(x) = 2^n(x^{1/2^n} - 1)$, $x > 0$, show that for each positive $x \neq 1$ the sequence $(L_n(x))$ is decreasing and has a finite limit and that $L_n(1) = 0$. Let $L(x) = \lim_{n \rightarrow \infty} L_n(x)$. Show by studying $L_n(xy) - [L_n(x) + L_n(y)]$ that L satisfies Eq. (5.1). Show that L is never decreasing, that is, $L(x) \leq L(y)$ for $x < y$, and conclude with the aid of Eq. (5.1) that L is continuous at 1 and thereupon on $\mathbf{P}$. [Consider $(L(a^{1/2^n}))$, $a > 1$.] Next, using the fact that $(1 + 2^{-n})^{2^n} < e$, $n \in \mathbf{N}$, but $\lim(1 + 2^{-n})^{2^n} = e$, e being $\lim[1 + (n+1)^{-1}]^{(n+1)}$, conclude

$$L[(1 + 2^{-n})^{2^n}] < L_n[(1 + 2^{-n})^{2^n}] < L_n(e),$$

and hence $L(e) \leq 1 \leq L(e)$, and so $L(e) = 1$. Now show that L is increasing, that is, $L(x) < L(y)$ when $x < y$.

Prove that L is the unique never decreasing (resp. continuous) map of $\mathbf{P}$ into $\mathbf{R}$ satisfying Eq. (5.1) and the normalization $L(e) = 1$. The function L is the *logarithm function* and will be denoted $\log$, as is customary. Show that $\log(\mathbf{P}) = \mathbf{R}$. Let $\exp$ denote the inverse of $\log$, which we term the *exponential function*. Conclude from the properties of $\log$ that $\exp$ is an increasing continuous map of $\mathbf{R}$ into $\mathbf{P}$ satisfying the addition theorem

$$\exp(x + y) = \exp(x) \exp(y), \quad x, y \in \mathbf{R}, \quad (5.2)$$

and $\exp(1) = e$. Show that $\exp$ is the unique real function with domain $\mathbf{R}$ satisfying Eq. (5.2) and $\exp(1) = e$, and being either continuous at some $a \in \mathbf{R}$ or never-decreasing.

5.2. Decimal approximants. Given $a \in \mathbf{R}^+$, $n \in \mathbf{N}$. Show that $\{p \in \mathbf{N}, 10^{-n}p \leq a\}$ has a maximum. We denote it $m(n)$. $10^{-n}m(n)$ is termed the *nth decimal approximant* of a . Show that $(m(n)10^{-n})_{\uparrow}$ has limit a .

† We speak of a *finite* limit in the latter case.

6. LIMIT SUPERIOR, LIMIT INFERIOR

The notions in question are very useful concepts in analysis. The notion of limit superior will play a role in Ch. II in the determination of the radius of convergence of a power series. While, at the very least, the larger part of the material of the preceding sections of this chapter should be familiar to students beginning the study of complex function theory, the notions of limit superior and limit inferior are not always known.

We start with a real sequence (a_n) and classify it according as $\{a_n: n \in \mathbb{N}\}$ is bounded above or not. We agree to say that (a_n) has limit superior $+\infty$, denoted " $\limsup_{n \rightarrow \infty} a_n = +\infty$," provided that $\{a_n: n \in \mathbb{N}\}$ is not bounded above. If $\{a_n: n \in \mathbb{N}\}$ is bounded above, so is $\{a_{m+n}: n \in \mathbb{N}\}$ for each $m \in \mathbb{N}$. In this case let $\alpha_m = \sup\{a_{m+n}: n \in \mathbb{N}\}$. We have $(\alpha_m) \downarrow$. We say that (a_n) has limit superior $-\infty$, denoted " $\limsup_{n \rightarrow \infty} a_n = -\infty$ " provided that $\{a_n: n \in \mathbb{N}\}$ is bounded above and $\lim_{m \rightarrow \infty} \alpha_m = -\infty$. We say that (a_n) has limit superior α where $\alpha \in \mathbb{R}$, denoted " $\limsup_{n \rightarrow \infty} a_n = \alpha$," provided that $\{a_n: n \in \mathbb{N}\}$ is bounded above and $\lim_{m \rightarrow \infty} \alpha_m = \alpha$.† Here we write $\limsup_{n \rightarrow \infty} a_n$ for α .

The corresponding introduction of the notion of limit inferior ($\liminf$) is the dual of that used above, "bounded above" being replaced by "bounded below," "sup" by "inf," " $+\infty$ " by " $-\infty$," " $-\infty$ " by " $+\infty$." The other modifications are obvious.

The first of the following exercises characterizes each of the situations considered in treating the notion of limit superior.

EXERCISES

6.1. $\limsup_{n \rightarrow \infty} a_n = -\infty$ if and only if $\lim_{n \rightarrow \infty} a_n = -\infty$. $\limsup_{n \rightarrow \infty} a_n = +\infty$ if and only if some subsequence of (a_n) has limit $+\infty$. $\limsup_{n \rightarrow \infty} a_n = \alpha \in \mathbb{R}$ if and only if for each positive η the set $\{n: a_n \geq \alpha + \eta\}$ is bounded and the set $\{n: a_n > \alpha - \eta\}$ is unbounded.

6.2. If $\limsup_{n \rightarrow \infty} a_n = \alpha \in \mathbb{R}$, then for every subsequence (b_n) of (a_n) such that (b_n) possesses a limit in $\mathbb{R}$ we have $\lim_{n \rightarrow \infty} b_n \leq \alpha$. Further, there exists a subsequence (b_n) such that $\lim_{n \rightarrow \infty} b_n = \alpha$. Hint: To construct such a (b_n) define $m(n)$, $n \in \mathbb{N}$, by $m(0) = \min\{k: |\alpha - a_k| < 2^0\}$, $m(n+1) = \min\{k: k > m(n), |\alpha - a_k| < 2^{-(n+1)}\}$, $n \in \mathbb{N}$, and show that $b_n = a_{m(n)}$ serves.

6.3. Given (a_n) bounded, that is, $\{a_n: n \in \mathbb{N}\}$ is bounded, then there exists a subsequence possessing a limit in $\mathbb{R}$.

6.4. Given $\alpha \in \mathbb{R}$ and (a_n) . Then $\lim a_n = \alpha$ if and only if $\limsup_{n \rightarrow \infty} a_n = \alpha$ and $\liminf_{n \rightarrow \infty} a_n = \alpha$. State and prove the corresponding result for $\lim a_n = +\infty$.

† In §6, Ch. III, we shall see that the notational devices employed above may be transformed in a unified way into meaningful assertions in terms of the compactification of $\mathbb{R}$ to be introduced there.

7. CAUCHY SEQUENCES

It is readily verified that if $\lim a_n = \alpha \in \mathbf{R}$, then the following condition, the so-called *Cauchy condition*, is fulfilled:

For each positive η there exists $m \in \mathbf{N}$ such that $|a_p - a_q| < \eta$ when $p, q \geq m$. (C)

It suffices to take m such that $|a_n - \alpha| < \eta/2$ when $n \geq m$. For then $|a_p - a_q| \leq |a_p - \alpha| + |\alpha - a_q| < \eta$ when $p, q \geq m$. The converse holds. It states an important property of $\mathbf{R}$. A *Cauchy sequence* is one that satisfies the condition (C). We have:

Every Cauchy sequence possesses a limit (in $\mathbf{R}$).

A particularly simple proof may be given with the aid of the concepts of limit superior and limit inferior. Let (a_n) be a Cauchy sequence. Invoking (C) for $\eta = 1$ we see that (a_n) is bounded. Let $\alpha_p = \sup\{a_{p+n} : n \in \mathbf{N}\}$ and let $\beta_p = \inf\{a_{p+n} : n \in \mathbf{N}\}$. The sequences (α_n) and (β_n) are bounded and monotone and hence possess limits in $\mathbf{R}$. Given positive η we see that from $a_{p+n} - a_q < \eta$, $p, q \geq m$ and $n \in \mathbf{N}$, we have $\alpha_p - a_q \leq \eta$, $p, q \geq m$, and hence $\alpha_p - \beta_p \leq \eta$ for $p \geq m$. Hence $\lim \alpha_p = \lim \beta_p$. We denote the common value by α . We assert that $\lim a_n = \alpha$. Indeed, given positive η , for some p we have

$$\alpha - \eta < \beta_p \leq \alpha_p < \alpha + \eta,$$

and hence

$$\alpha - \eta < a_n < \alpha + \eta, n \geq p.$$